

**Deciphering cell–cell communication
at single-cell resolution for spatial transcriptomics with
subgraph-based graph attention network
(Yang et al., *Nature comm* (2024))**

Jounal Meeting

2024. 09. 24

변가은

KOREA
UNIVERSITY

Overview

1. Introductions

- Backgrounds
- DeepTalk

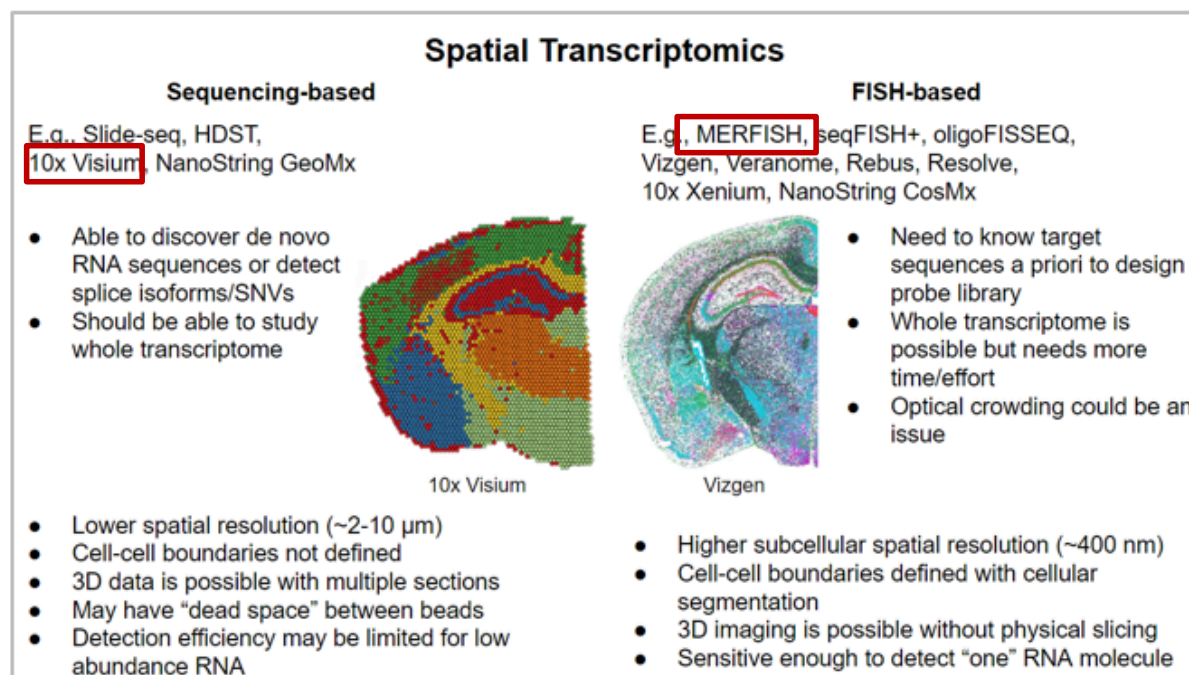
2. Structure of DeepTalk and Benchmarking

- DT-Integration
- DT-CCC

3. Evaluation of DeepTalk with 10x Visium Data sets

4. Discussion

Spatial Transcriptomics



Technology	Type	Target	Resolution (nm)
MERFISH	FISH	RNA	100
GeoMx	NGS	RNA, protein	10,000
Visium	NGS	RNA	55,000
Stereo-Seq	NGS	RNA	500
CosMx	FISH	RNA, protein	50
Xenium	FISH	RNA	50
Visium HD	NGS	RNA	5,000

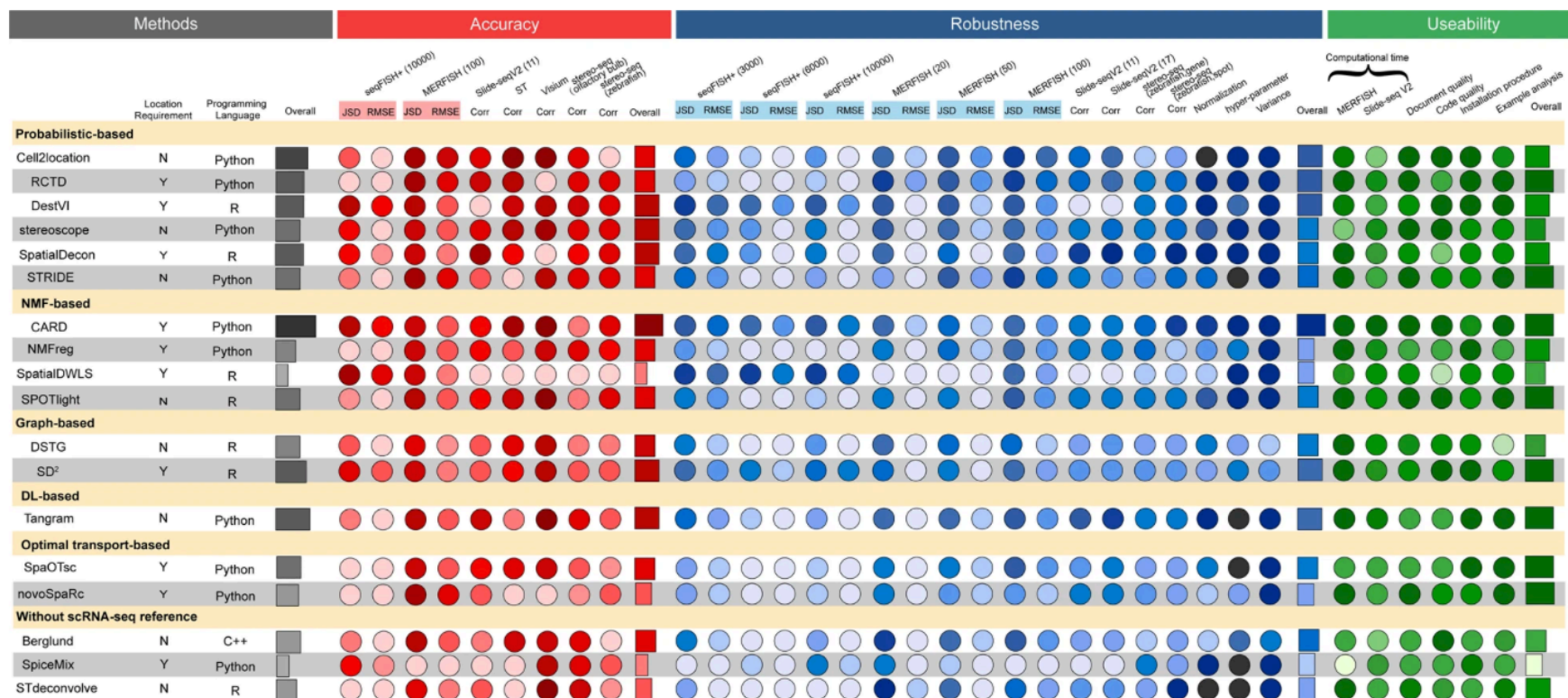
- Gene expression + spatial information(position...)
 - Accurate Cell Composition, Cell-Cell Interaction

- “**Spot**” containing several cells
 - Deconvolution : Reconstruction of Cell distribution

Spatial Transcriptomics

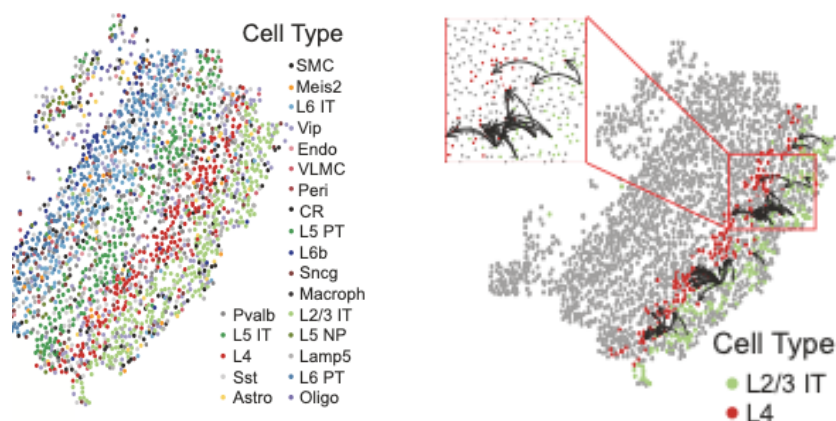
- **Limitation of Spatial Transcriptomics Analysis**

1. 낮은 Spot-resolution으로 인한 정확한 Spatial Cell Type Annotation 어려움
2. 전체 RNA를 capturing하지 못함



Introducing DeepTalk

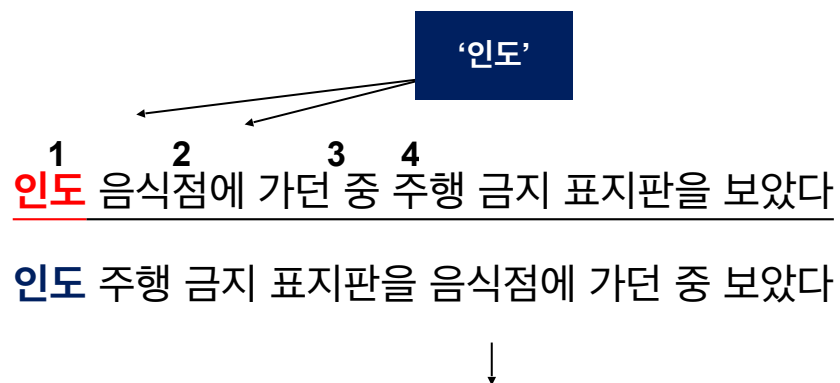
- Using **Graph Attention Network**
 - Self-attention & Cross-Attention
- Predicting more accurate cell type & cell-cell communication(CCC)
 1. **Cell type annotation or Deconvolution** by Integrating ST data + sc/scRNA-seq data
 2. **Cell-Cell Communication** considering spatial context



Discovering
Fine-grained tissue organization,
Cellular-level communication

Graph Attention Network

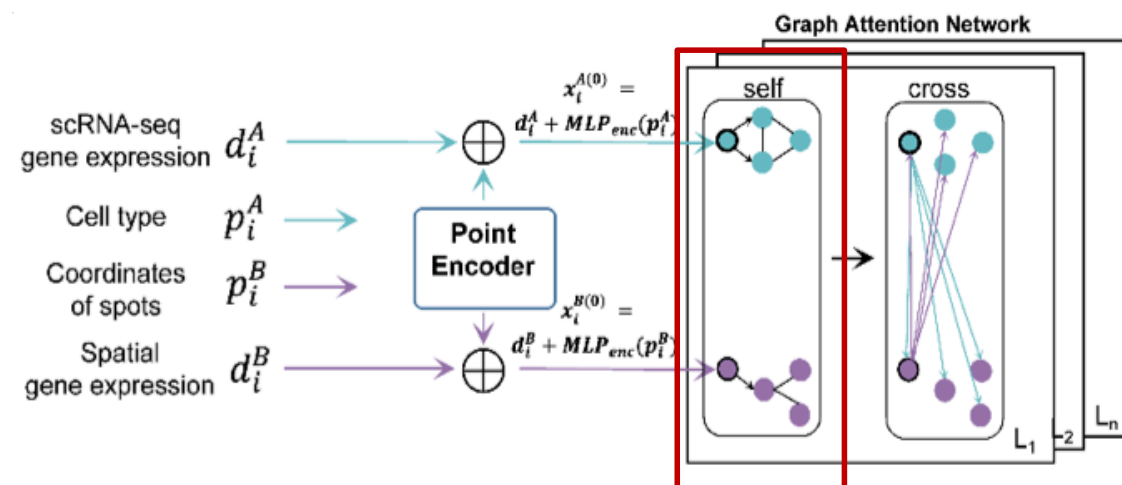
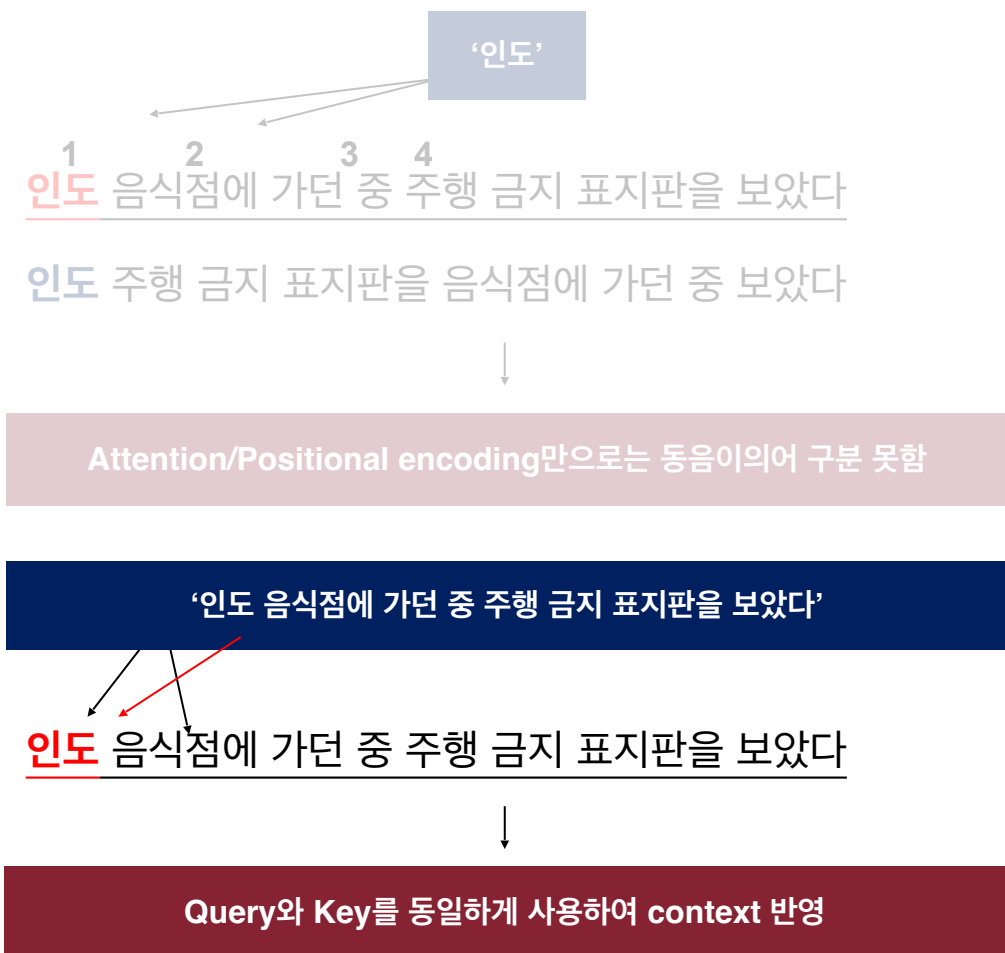
- Self-Attention



Attention/Positional encoding만으로는 동음이의어 구분 못함

Graph Attention Network

- Self-Attention



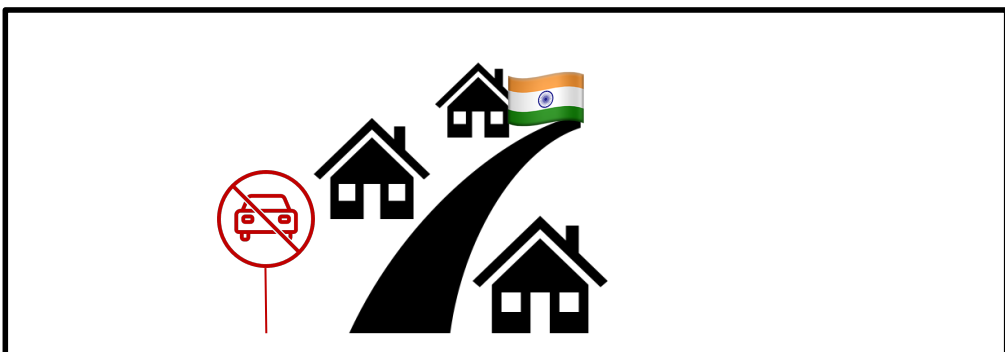
- Point Encoder** : Cell type과 Spot 위치를 encoding
- Self-Attention** : 주위 cell들의 expression을 반영

Graph Attention Network

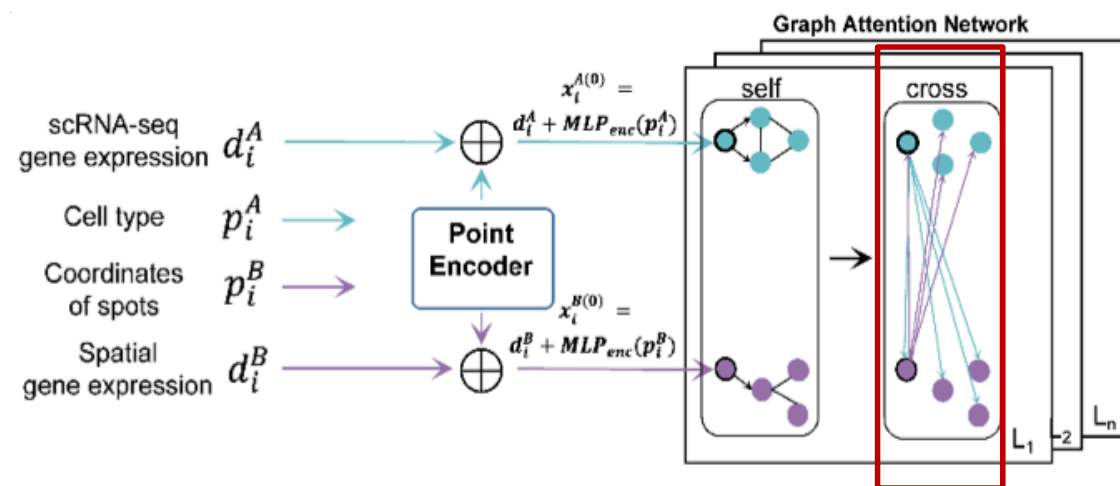
- Cross-Attention

Q
:
K
:

‘인도 음식점에 가던 중 주행 금지 표지판을 보았다’



Multi-modal data를 sequence로 사용하여 embedding

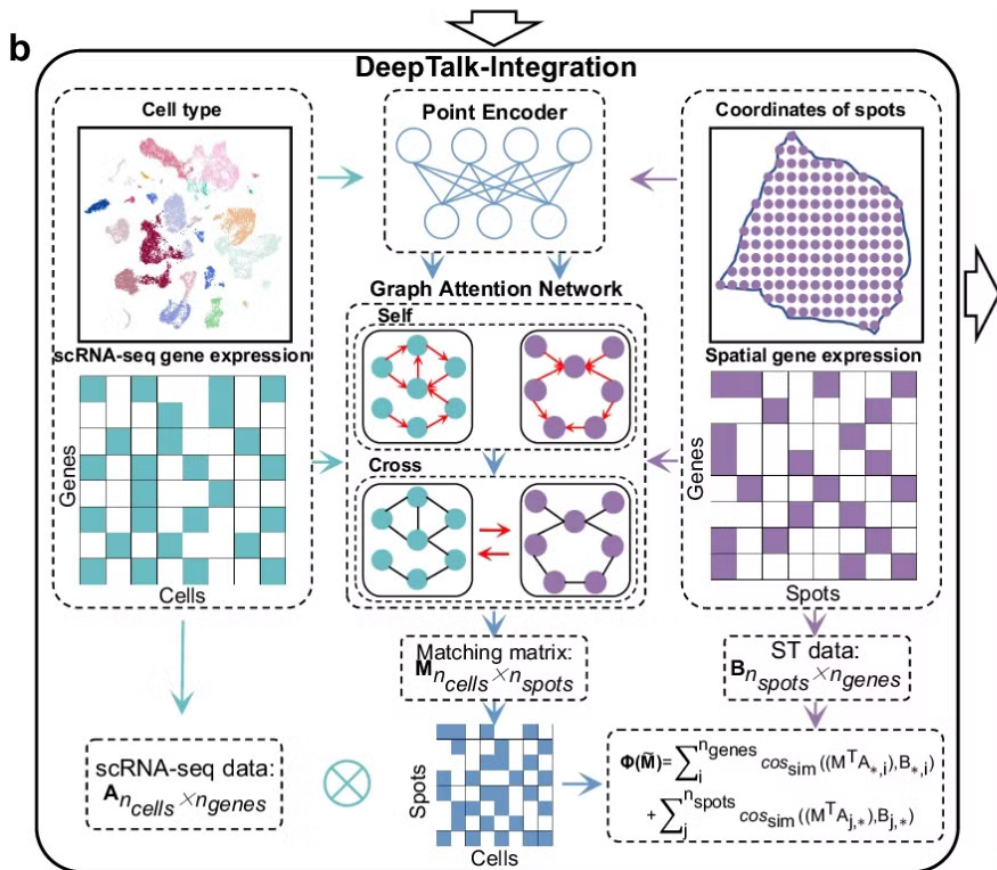


- Cross-Attention** : sc/snRNA-seq data와 ST data의 connection 반영

DeepTalk- INT & DeepTalk- CCC

DeepTalk-Integration

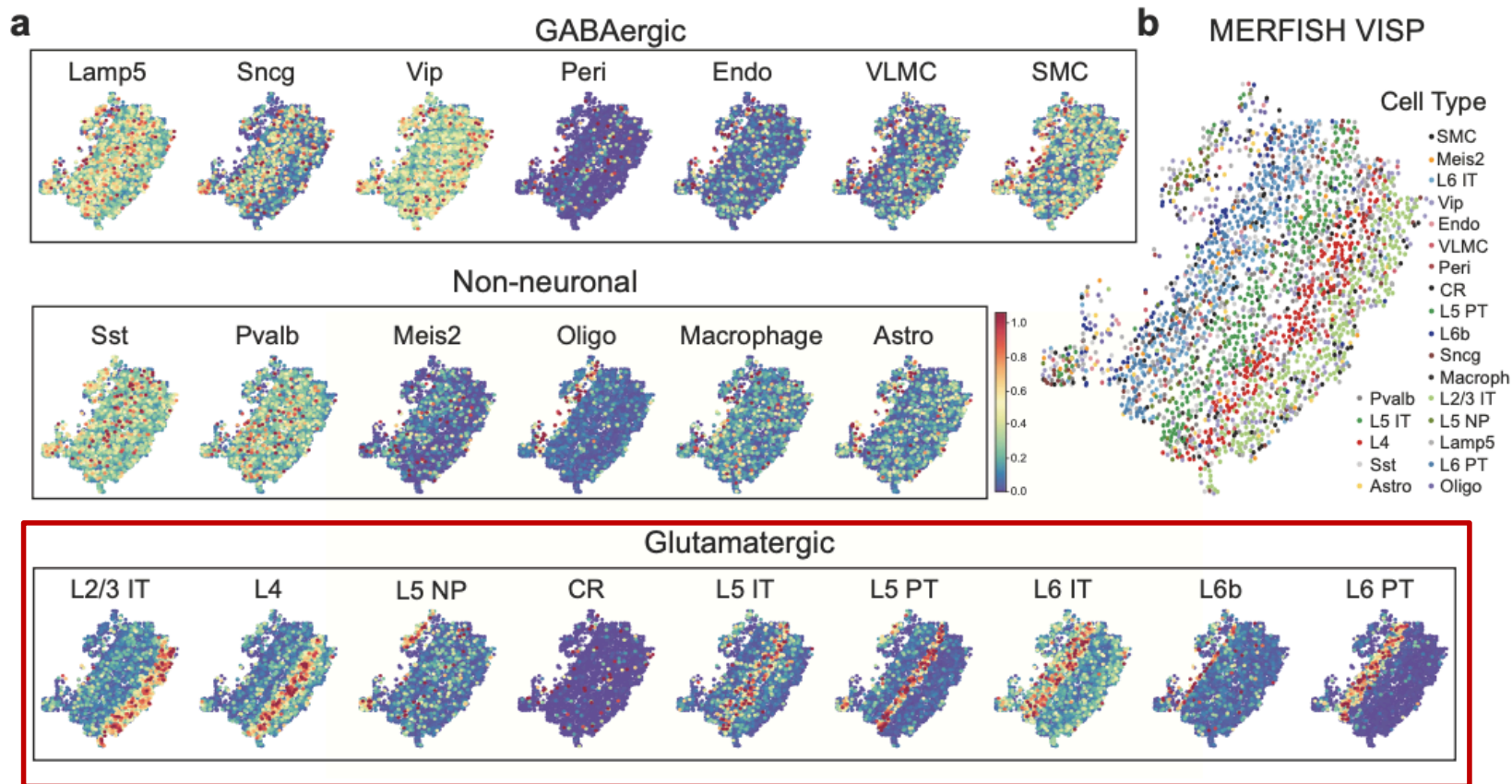
: focusing on **determining the cell-type composition of sc or spot ST datasets**



1. sc/snRNA-seq data와 ST data matrix를 **decoding**
2. Weight matrix를 형성
 - Cell(sc-based ST) or spot(spot-based ST)에 대한 weight
3. Integrated data에서 **max.weight**를 가지는 **cell type**으로 **labelling**
(spot-based: 여러 cell type의 weights info.로 iteration을 통해 cell combination 계산해 sc수준의 spatial map 형성)

DeepTalk-Integration

Cell type annotation of Mouse visual cortex (MERFISH)

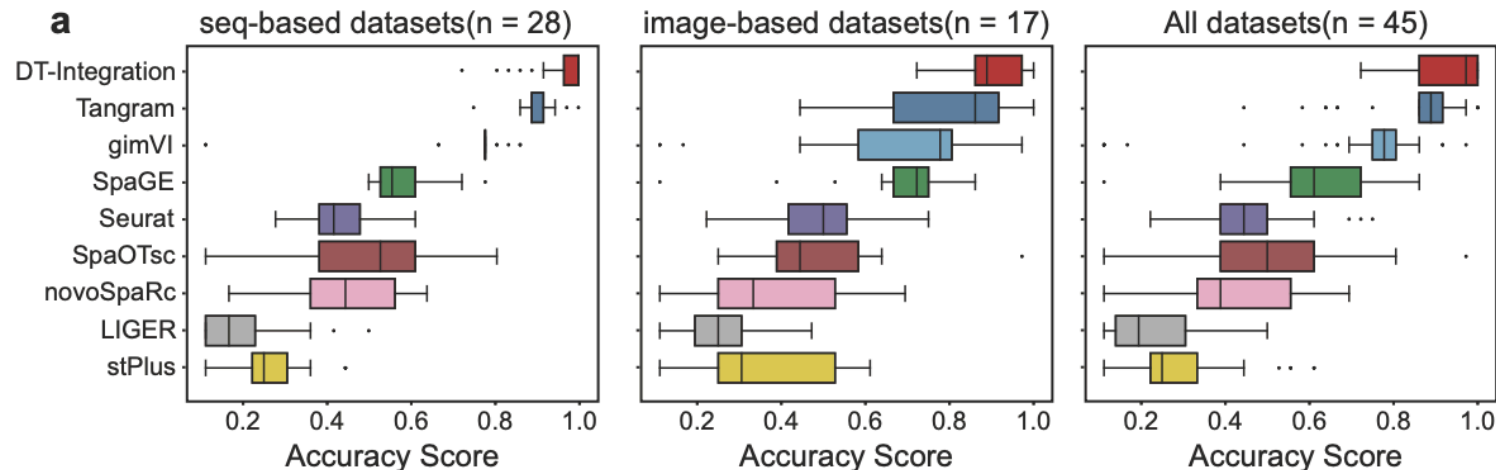


✓ Unique patterns along cortical layers of Glutamatergic cells

DeepTalk-Integration

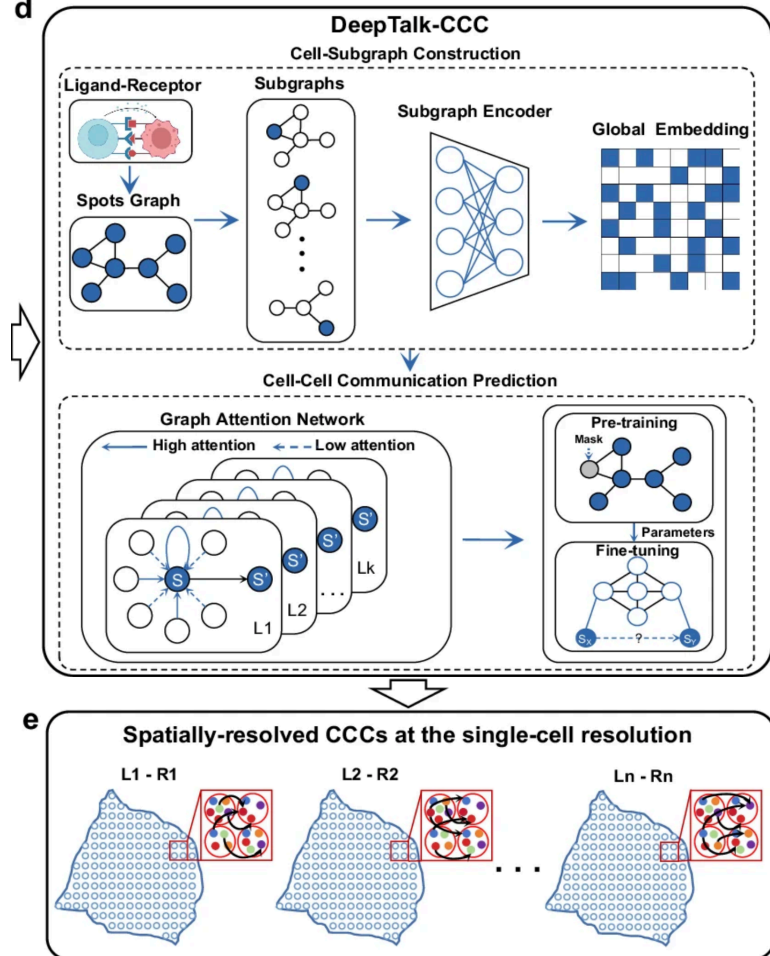
- “Integrated data가 ST data에서 detecting하지 못한 transcripts의 spatial dist.을 얼마나 잘 예측하나?”
 - With Tangram, gimVI, SpaGE, Seurat, SpaOTsc, novoSpaRc, LIGER, stPlus
 - Metrics
 - : **PCC**(Pearson correlation coefficient), **SSIM**(Structure similarity index), **RMSE**(Root mean square error), **JSD**(Jenson-Shannon divergence)

$$AS = \frac{1}{4}(\mathbf{RANK}_{PCC} + \mathbf{RANK}_{SSIM} + \mathbf{RANK}_{RMSE} + \mathbf{RANK}_{JS})$$

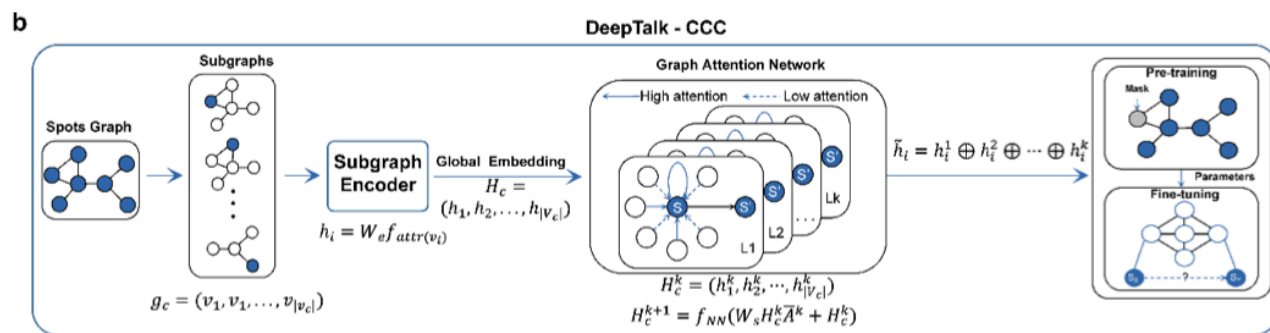


DeepTalk-CCC

: focusing on **predicting the spatially mediated CCCs influenced by Pairs(L-R pairs) of genes in the spatial context**



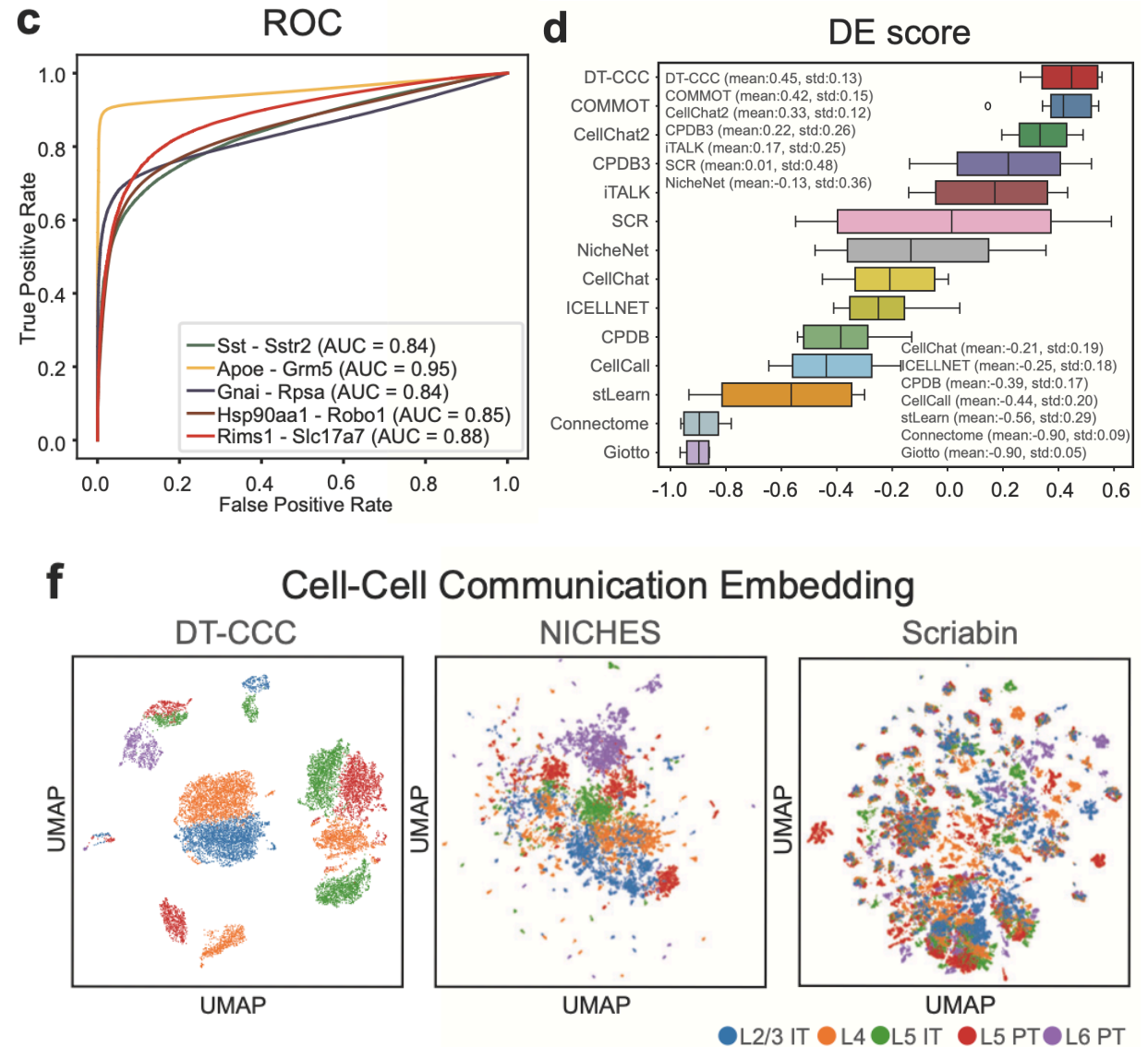
1. 각 Cell의 local feature에 대한 **sub-graph encoding**
 - Subgraph encoder : 벡터 형태로 변환하여 GAT에서 처리할 수 있도록 함
2. Sub-graph node(cell)의 중요도를 계산 (**GAT**)
 - High/low attention 학습



➤ pre-training or fine-tuning

DeepTalk-CCC

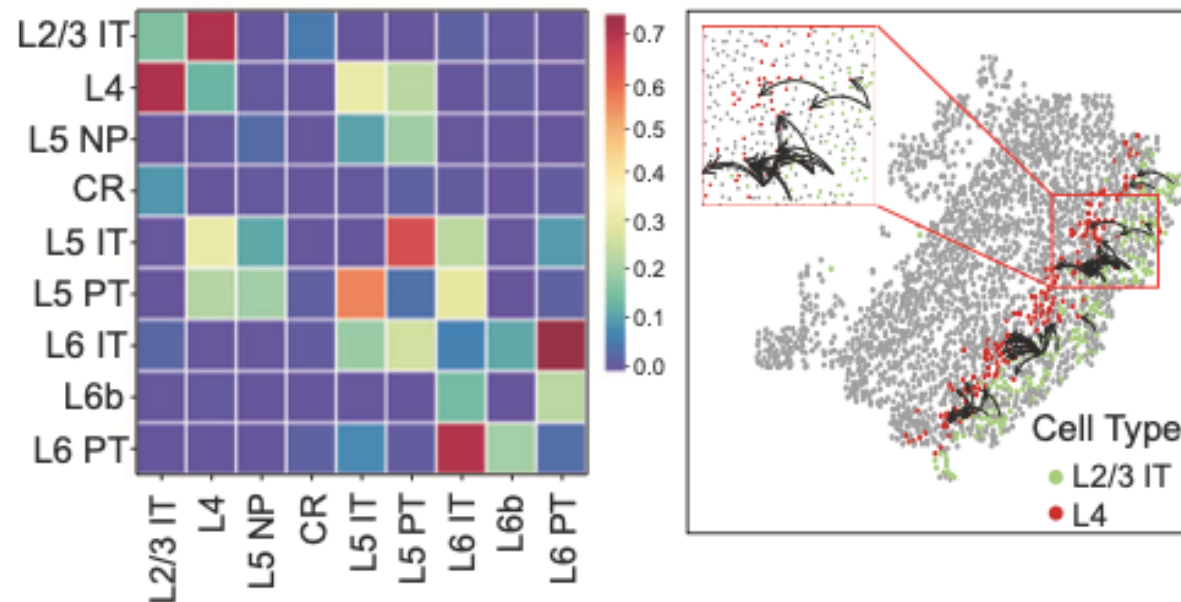
- With MERFISH dataset
 - ROC : 0.84~0.95 (accuracy)
 - DES(The distance enrichment score)
 - : 0.45 (uniformity)



DeepTalk-CCC

- With MERFISH dataset
- Apoe-Grm5
 - : modulates the activation pattern of Grm5, affecting glutamatergic signal transmission and neuronal activity

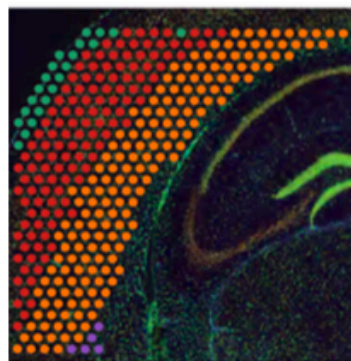
i CCCs mediated by Apoe - Grm5 pairs



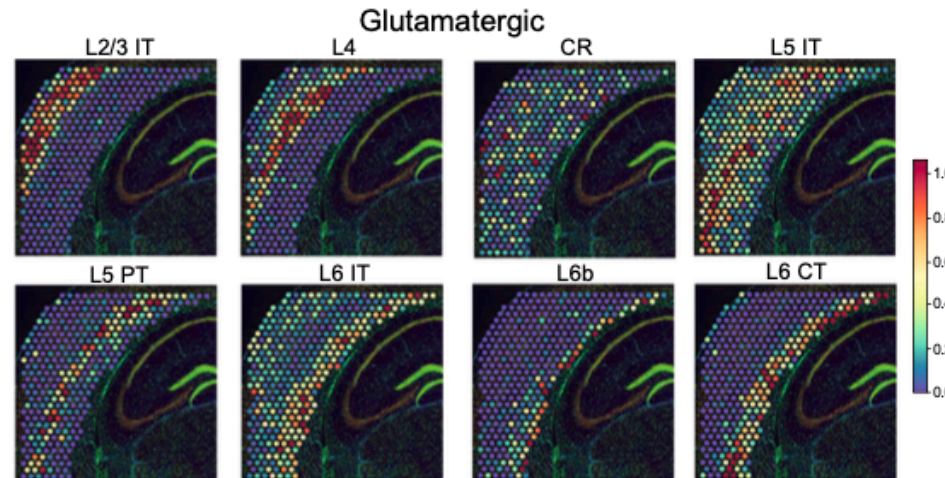
Single-cell level delineation in adult mouse brain

- Training a model with 10x Visium(55μm)
+ scRNA-seq data(1,291 genes from postnatal day 2 and 11 mouse brains and spinal cords)
- Applied to adult mouse cortical region scRNA-seq data(21,697 cells)
 - Preprocessed with Scanpy, Squidpy(ensuring an equal number of segmented slices per cell, so that enable the precise mapping of each cell within each voxel, thereby **predicting cellular localization at a single-cell resolution**)

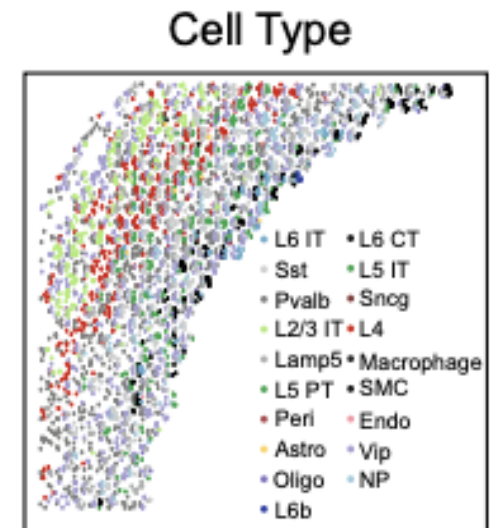
a Probability mapping
Adult Mouse Brain
10x Visium



Cluster
Cortex_1
Cortex_2
Cortex_3
Cortex_4



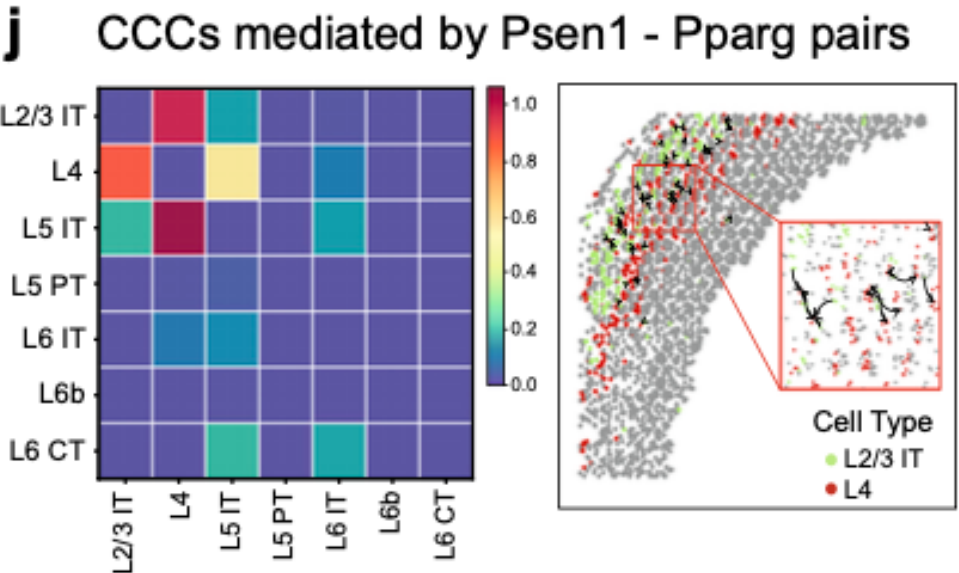
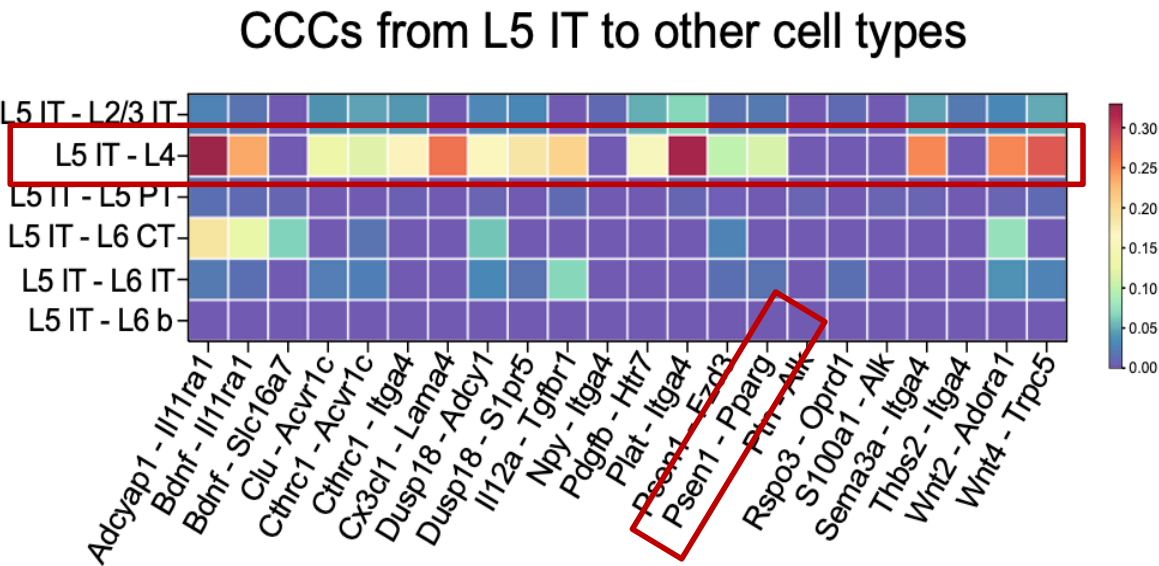
DT-Int
→



Single-cell level delineation in adult mouse brain

predicting CCC between L5 IT – other layers by glutamatergic cells

- **Psen1** (Presenilin 1) is intricately linked to Alzheimer’s disease and fulfills crucial functions in intracellular signal transduction as well as membrane protein transport.
- Glutamate-facilitated communication among cells essentially depends on the meticulous release and reception of the neuro- transmitter glutamate, a process that is meticulously modulated by a range of factors, including intracellular signaling cascades and the transcriptional activities of regulatory factors.



Discussion

Discussion

DeepTalk...

- utilizes an **attention mechanism-based GNN** to accurately predict the L–R pairs that mediate intercellular communication and visualize CCC at multiple scales
- can **visualize** the spatial communication mediated by the enriched intercellular L–R pairs
- **long-distance communication**
- **multimodal data amalgamation**